Pediatric Neurology 142 (2023) 76–88

Contents lists available at ScienceDirect

Pediatric Neurology

journal homepage: www.elsevier.com/locate/pnu

Topical Review

The Impact of Screen Exposure on Attention Abilities in Young Children: A Systematic Review

Marie Jourdren, MD [a, *], Aurélie Bucaille, PhD [a, b], Juliette Ropars, MD, PhD [a, c, d]

[a] CHRU de Brest, Service de pédiatrie, Brest, France
[b] Centre de Référence des Troubles des Apprentissages, CHU de Brest, Brest, France
[c] Laboratoire de Traitement de l'Information Médicale INSERM U1101, Brest, France
[d] Groupe d'Intérêt Scientifique Be A Child, Brest, France

ARTICLE INFO

Article history:
Received 31 March 2022
Accepted 10 January 2023
Available online 16 January 2023

Keywords:
Screen exposure
Attentional functions
Preschool children
Systematic review

ABSTRACT

Background: The neurocognitive consequences of exposure to screens in school-aged children have been the subject of multiple studies. However, the relationship between screen exposure and neurocognitive development, especially attentional functions, remains unclear in preschool children. The aim of this systematic review was to evaluate the immediate and long-term impact of screen exposure on attentional functions in preschoolers.
Methods: Three electronic databases (PubMed, COCHRANE, and SCOPUS) were searched for studies published between January 1, 2000, and November 30, 2020. Two reviewers independently selected studies. Inclusion criteria were observational studies, inclusion of children aged less than seven years with no neurodevelopmental disorders, evaluation of screen time, and evaluation of attentional functions. Data extracted including participants' ages, number of participants, screen exposure time, attention assessment tool, and confounding factors. Study quality was evaluated using the Newcastle-Ottawa Scale.
Results: Five cross-sectional studies were included: all reported significant, positive associations between high levels of screen exposure and attention difficulties. Ten longitudinal studies were included: six found a significant impact of earlier screen exposure on subsequent attentional function and four found no relationship. Eight of the studies included evaluated the direction of the relationship between screen exposure and attentional difficulties: seven suggested the relationship is bidirectional.
Conclusions: These findings support current guidelines to limit screen time in preschoolers to prevent the development of attention difficulties.

© 2023 Elsevier Inc. All rights reserved.

Introduction

Children, including preschool children (defined here as children aged six years or younger), are widely exposed to screens in their daily lives: studies performed in the last five last years showed that in developed countries each household owns on average seven screen devices.[1,2] The mobility of digital media makes their presence ubiquitous. Screen exposure is defined as the time spent looking at a screen either through a fixed device, such as a television, or a mobile device, such as a smartphone or tablet, anytime and anywhere throughout the day.[3]

Already 20 years ago, screen use by children was high. A study in 2003 showed that 68% of children younger than two years used screen media daily, with an average screen time of 2.05 hours per day.[4] More recent surveys found similar rates in children aged zero

Declarations of interest: None.

Clinical Trial Registration: Registry name: PROSPERO, registration number: CRD42021238488; web link to study on registry: https://www.crd.york.ac.uk/PRweOSPERO/.

Contributors' Statement: Dr. Bucaille critically reviewed and revised the manuscript for important intellectual content. Dr. Jourdren conceptualized and designed the study, drafted the initial manuscript, was one of the reviewers for article selection, and critically reviewed and revised the manuscript. Dr. Ropars conducted the search, designed the study, designed the data collection instruments, was one of the two reviewers for article's selection, and critically reviewed the manuscript for important intellectual content. All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

* Communications should be addressed to: Dr. Jourdren; Service de pédiatrie; CHRU de Brest; 2 avenue Foch; Brest Cedex 29609, France.
E-mail address: marieJourdren@hotmail.fr (M. Jourdren).

https://doi.org/10.1016/j.pediatrneurol.2023.01.005
0887-8994/© 2023 Elsevier Inc. All rights reserved.

to six years.[5,6] This level of exposure far exceeds current guidelines: the American Academy of Pediatrics (AAP) recommends that toddlers aged below 18 to 24 months should not have access to digital media (except for video chatting). The AAP also recommends limiting screen use in preschoolers (i.e., those aged two to five years) to one hour per day of high-quality programming.[7] The French Academy of Sciences recommends avoiding screens before age 24 months.[1]

A qualitative study of 25 parents found four main motives for parents to expose their children to screen media: to entertain them, to calm them, to recompense them, and to educate them.[8] However, studies have shown that screen exposure can negatively impact many aspects of a child's life. In older children and teenagers, excessive screen exposure is associated with adverse somatic consequences such as overweight,[9,10] impaired sleep,[11-13] high blood pressure,[14] and metabolic disorders.[15] Adverse associations have also been found on neurocognitive development including poor cognitive outcomes[13,16] and low academic performance.[17,18] In younger children, excessive screen use has been found to impact social interactions between children and to slow the development of language skills.[19-22]

Attentional skills develop from a very early age, and different brain networks have been found to contribute to the three main attention functions: alerting emerges at birth, orienting matures rapidly within the first six months, and executive control starts to become established toward the end of the second year.[23,24] The attention system is the foundation for the development of many functions including cognitive functions, such as execution[25] and learning,[26,27] and social functions.[28] Maturational processes of executive systems are protracted into adolescence and beyond. The impact of attentional skills on neurocognitive development carries over into adulthood.[29] Despite this, the impact of high screen exposure on attentional functions has been little studied and the consequences remain unclear. Advancing early childhood development is a public health priority,[30] thus research to improve understanding of attentional functions and the impact of screen exposure on their development is important.

We therefore conducted a systematic review of the literature to assess the impact of screen exposure on attentional functions in preschoolers. The specific questions were as follows.

(1) Is there an immediate association between the quantity of screen exposure and attentional function (i.e., at the same age) in preschoolers?
(2) Does early screen exposure have a long-term impact on subsequent attentional functions?
(3) Does screen use influence attentional function or does attentional function influence screen use?

Methods

This systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis statement.[31]

The review protocol was registered on the International Prospective Register of Systematic Reviews (reference number CRD42021238488).

Data sources

Articles were identified through a systematic search of the PubMed, Scopus, and Cochrane Library databases. We selected articles that were published between January 1, 2000, and November 30, 2020. The year 2000 was chosen because a collective awareness of the effects of screen exposure began around this time and the first official recommendations on the subject were published in 1999 by the AAP.[32]

The search string was composed of Medical Subject Heading terms and text words, with different combinations of keywords to achieve exhaustivity. The search terms used included the following concepts: “screen time” (words used: screen time, screen media, media, television, video games, cell phone), “attention” (words used: attention, ADHD), and “children” (words used: child, children, infant, preschooler, toddler). Synonymous terms were first associated with the Boolean “OR.” Concepts were then combined with the Boolean “AND” (Supplemental Appendix 1).

Study selection

Studies had to meet the following criteria to be included: (1) full original study published in English; (2) sample composed of at least 30 subjects; (3) children aged seven years or less (or subgroups with children younger than seven years that could be clearly identified); (4) measurement of attention capacities; and (5) quantitative measurement of screen exposure time. Exclusion criteria were the following: (1) studies that included children with neurodevelopmental disorders such as autism, attention-deficit/hyperactivity disorder (ADHD), or neurological conditions such as brain injury; (2) experimental or interventional studies; and (3) conference abstracts, case reports, and letters to the editor.

Two reviewers (M.J. and J.R.) independently screened and assessed the studies for eligibility. Articles were classified according to their titles and abstracts as potentially relevant or definitely irrelevant. The full texts of studies classified as potentially relevant were reviewed and included if they met the inclusion criteria. The references of included articles were searched to identify any other potential studies. The reviewers’ final selections were compared, and any discrepancy was discussed until a consensus was reached.

Data extraction

The reviewers independently extracted data from the studies included into an extraction template using a standardized form. The following information was extracted from each study: year of publication, location of study, design, duration of observation (for longitudinal studies), sample size, age range, sex, confounding factors, measures of screen exposure, measures of attentional function, and main outcomes. Associations between screen exposure and attention were exhaustively reported.

The source material included was very heterogeneous with regard to the populations studied, the tools used to measure attention, and the statistical methods used. As a result, pooled analysis could not be carried out. We therefore performed a qualitative synthesis of the studies included, according to the three research questions addressed by the systematic review.

Risk of bias assessment

All studies were independently assessed by the two authors using the Newcastle-Ottawa Scale (NOS) since all the studies selected were observational.[33] For cohort studies (Table 1), the NOS is based on the following criteria: selection bias (maximum score: 4), comparability of study groups (maximum score: 2), and ascertainment of exposure for case-control studies or outcome for observational studies (maximum score: 3). Longitudinal studies were considered to have a low risk of bias if they scored $\geq$8 and a moderate-high risk of bias if they scored <8. For cross-sectional studies (Table 2), a score for quality, modified from the NOS, was used.[43] The NOS adapted for cross-sectional studies assessed

TABLE 1
Results of Risk Bias Assessment for Longitudinal Studies (Newcastle-Ottawa Scale)

Cohort Studies	Overall Risk of Bias Score	Criteria								
		Selection				Comparability		Outcome		
		Representativeness of the Exposed Cohort	Selection of the Nonexposed Cohort	Ascertainment of Cases	Outcome of Interest Was Not Present at Start of Study	Adjust for the Most Important Risk Factors	Adjust for Other Risk Factors	Assessment of Outcome	Sufficient Follow-up Duration	Adequacy of Follow-up and Cohorts
Cheng et al.[34]	8	x	x	x	x	x	-	x	x	x
Chonchaiya et al.[35]	7	-	x	x	x	x	-	x	x	x
Christakis et al.[36]	9	x	x	x	x	x	x	x	x	x
Gueron-Sela and Gordon-Hacker[37]	7	x	x	x	x	x	-	x	x	x
Parkes et al.[38]	8	x	x	x	x	x	-	x	x	x
Peralta et al.[39]	8	x	x	x	x	x	-	x	x	x
Stevens and Muslow[40]	7	x	x	x	-	x	-	x	x	x
Tamana et al.[35]	8	x	x	x	x	x	-	x	x	x
Tomopoulos et al.[41]	6	-	x	x	.	x	-	x	x	x
Verlinden et al.[42]	8	x	x	x	x	x	x	-	x	x

x: An individual criterion is satisfied.
-: An individual criterion is not satisfied.
Low risk of bias: overall risk of bias score ≥8.
High risk of bias: overall risk of bias score <8.

appropriateness of research design, recruitment strategy, response rate, representativeness of sample, objectivity/reliability of outcome determination, power calculation provided, and appropriate statistical analyses. The maximal score is 10. There is no consensus definition of high or low risk of bias.[43] In this systematic review, studies scored ≥8 were considered to have a low risk of bias.

Results

Study selection

The initial search identified 1029 articles (125 from Cochrane, 202 from PubMed, 702 from Scopus). After duplicates were removed and titles were selected, 156 articles were classified as potentially relevant. After reading the abstracts, 42 articles were considered as potentially eligible and underwent full-text review. Of these, 30 did not fulfill the inclusion criteria. Three additional articles were identified from review of the references; therefore a total of 15 studies were included (Fig).

Study characteristics

The main characteristics of the selected studies are described in Table 3. Of the 15 studies, five were conducted in North America, seven in Asia, and three in Europe. Sample sizes varied widely across studies, ranging from 99 to 11,014 children, and the ages ranged from six months to seven years. The proportion of males to females was around 50% (46% to 61.8%). In only one study[39] children were extracted from a subgroup of children aged seven years or less. Confounding factors, such as gestational age, birth weight, family economic status, and social status, were taken into account in the subject selection or statistical analysis in the vast majority of studies. Five studies were cross-sectional and 10 were longitudinal.

The risk of bias is presented in Table 1 (using the NOS 33 for longitudinal studies) and Table 2 (using the modified NOS[43] for cross-sectional studies).

Immediate impact of screen exposure on attentional functions

All five cross-sectional studies included in this systematic review reported significant, mild to moderate positive associations between screen exposure and attention difficulties at the same age.

Ozmert et al.[45] and Xie et al.[46] found a positive correlation between the attention problem score (evaluated using the Child Behavior Checklist [CBCL]) and time spent watching television, with, respectively, $r = 0.2$ ($P < 0.001$) and $t = -5.15$. Similarly, Miller et al.[47] found that television exposure accounted for a significant proportion of the variance of inattentive/hyperactive behaviors ($r^2 = 0.168$ [$P = 0.02$]) in teacher ratings of behavior.

TABLE 2
Results of Risk Bias Assessment for Cross-sectional Studies (Newcastle-Ottawa Scale Adapted for Cross-sectional Studies)

Cohort Studies	Overall Risk of Bias Score	Criteria						
		Selection				Comparability	Outcome	
		Representativeness of the Sample	Sample Size	Nonrespondents	Ascertainment of the Exposure	Comparability of the Design and Analysis	Assessment of Outcome	Statistical Test
Hosokawa and Katsura[48]	10	x	x	x	x	x	x	x
Lin et al.[44]	8	x	x	x	-	x	x	x
Miller et al.[47]	7	-	x	x	x	x	x	x
Ozmert et al.[45]	7	-	x	x	x	x	x	x
Xie et al.[46]	9	x	x	x	x	x	x	x

x: An individual criterion is satisfied.
-: An individual criterion is not satisfied.

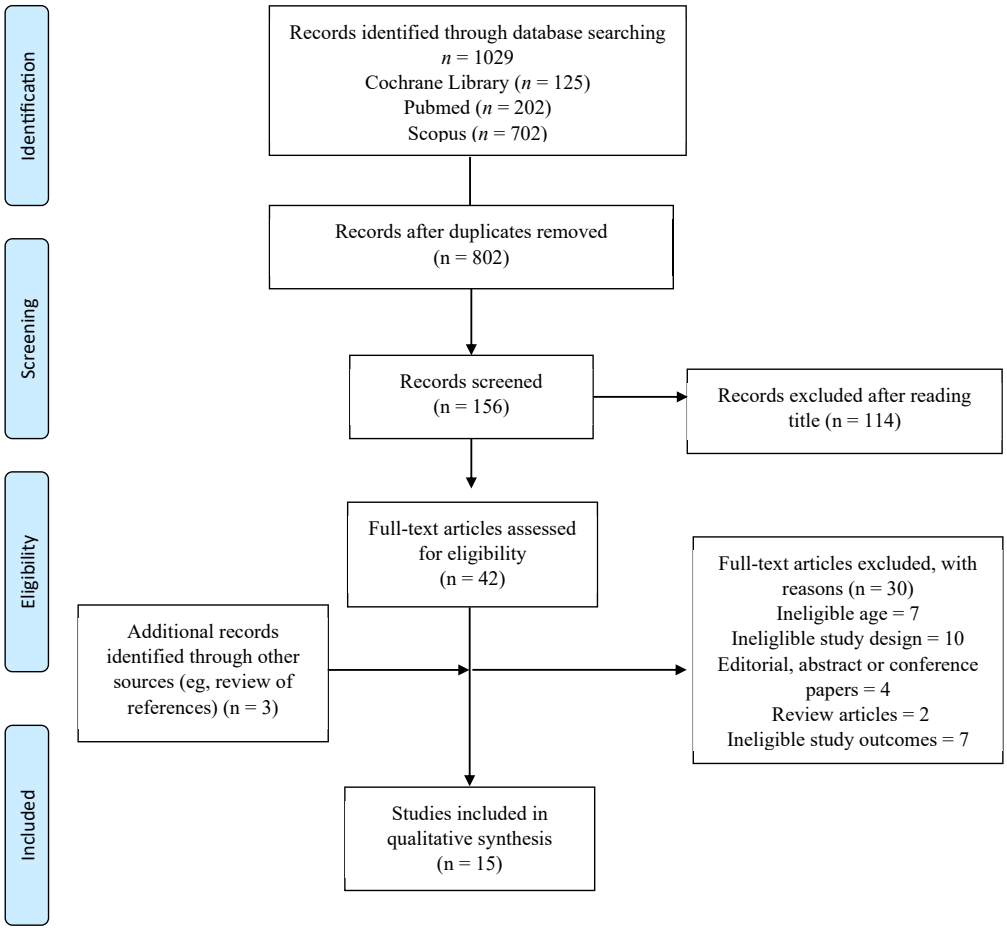

FIGURE. Preferred Reporting Items for Systematic Reviews and Meta-Analyses flowchart of the study selection. The color version of this figure is available in the online edition.

Two studies focused on miscellaneous uses of screens. Hosokawa and Katsura compared nonregular (<1 h/day) and regular use (>1 h/day) of mobile devices and found a significant association between regular use and higher hyperactivity/inattention scores using the Strengths and Difficulties Questionnaire (SDQ), with an inverse probability-of-treatment weighted odds ratio (OR) 1.82, 95% confidence interval (CI), [1.15 ± 2.87], $P < 0.01$.[48] Lin et al. found a positive association between prolonged touch screen usage and attention difficulties (CBCL) r = 0.3, $P < 0.001$.[44]

Long-term impact of screen exposure on attention abilities

The 10 longitudinal studies included evaluated the long-term association between screen exposure at a young age and attention difficulties months or years later, and six found a positive association. Christakis et al. found a proportional impact of the number of hours of television viewed per day at ages one and three years on the Behavioral Problems Index at seven years, with respective ORs of 1.09 (CI, 1.03 to 1.15) and 1.09 (CI, 1.02-1.16).[36] Cheng et al.,[34] Gueron-Sela and Gordon-Hacker,[37] and Verlinden et al.[42] studied screen exposure at 18 months and attention at, respectively, 30 months, 22 to 26 months, and 24 months. All these studies found a positive association between exposure and higher levels of attentional problems (rated using the SDQ subscales, Cumulative Media Use, and CBCL). Toumopoulos showed that media exposure at 21 months was associated with externalizing problems (OR = 1.7, $P = 0.046$) at 21 and 33 months.[41] Tamana et al. compared children aged five years who watched screen media for more than two hours per day or less than 30 minutes per day when they were aged three years. The authors showed that children exposed to greater screen times had higher inattention problem scores (CBCL) (OR, 5.9; 95% CI, 1.6% to 21.5%; $P < 0.01$).[35]

Four studies found no significant association between screen time and attentional functions. Stevens and Muslow found effect sizes for the relationship between television exposure on later symptoms of ADHD that were close to zero and not statistically significant.[40] The studies by Peralta et al. and Parkes et al. also found that high levels of screen exposure in children aged four 47 or five years[38] did not increase inattention symptoms at seven years (evaluated using Conner's Parent Rating Scale and the SDQ). Chonchaiya et al. found a nonsignificant (although positive) effect of background television exposure from six to 18 months on CBCL scores at 18 months.[49]

Direction of the relationship

Only one study assessed the direction of the relationship of screen exposure and attention.[37]

This medium-quality study found a significant negative indirect path between cumulative media used at 18 months and child focused attention at 26 months ($\beta = -0.12$, $P = 0.003$ [95% CI, −0.20 to −0.04]). Notably, the path between child focused attention and media used at 26 months showed a nonsignificant trend ($\beta = -0.13$, $P = 0.079$ [95% CI, −0.28 to 0.01]).

TABLE 3
Description of the Studies Included

Author, Year, Country	Study Design	Study Population	Covariates	Ascertainment of Exposure	Ascertainment of Outcome	Outcome(s)
Gueron-Sela et Gordon-Hacker, 2020, Israel[34]	L P	N = 302	-Sex	Question "How much TV the child watched daily"	The Japanese version of SDQ	Significant association between hours of watching TV at 18 mo (classified as "high group") and hyperactivity-inattention problems at 30 mo. F = 3.03, P = 0.029
		Healthy infants 50.7 % Male	-Birth weight -Gestational age	At age 18 and 30 mo Completed by mother	Mean scores of each SDQ subscale, ranging from 0 to 10	
		Age: 18 mo and then 30 mo	-Number of children in family	4 groups		
		Multicentric study	-Maternal educational level	<1 h/d	At age 30 mo.	
		From Japan Children's Study (prospective developmental cohort study)	-Family income	≤1 to <3 h/d	Completed by mother	
			-Level of mother's cognitive stimulation	≤3 to <4 h/d ≥4 h/d		
Chonchaiya et al., 2015, Thailand[38]	L P	N = 194	-Sex	24-hour media diary	The CBCL/1½-5	Adjusted r^2 total duration of TV exposure (min) at 18 mo on the CBCL behavioral scores: 0.085, P = 0.057
		Healthy infants	-Birth order	At age 6 and 18 mo	Raw scores of subscales	
		50.5 % Male	-Primary caregiver	Completed by mother	T-scores of internalizingand externalizing behaviors and total problem scores	Adjusted r^2 for increasing duration of TV exposure (min) from 6 to 18 mo on the CBCL behavioral scores at 18 mo.: 0.085, P = 0.138
		Mean age: 6.28 mo (S.D. 0.21) Mean age: 18.31 mo. (S.D. 0.30)	-Mother's and father's education in y -Family income	Name and duration of each TV program showing while the infant was awake	At age 18 mo	
		Monocentric study	-Child's temperament	Total duration of TV exposure, both foreground and background TV, per day	Completed by mother	
Christakis et al., 2004, US[44] Gueron-Sela et Gordon-Hacker, 2020, Israel[34]	L R	At 1 y: N = 1278	-Sex	Semistructured interview	Scale derived from BPI	Hours of TV viewed per day at both ages: at 1 y was associated with attentional problems at 7 y: OR 1.09 [1.03–1.15]; at 3 y was associated with attentional problems at 7 y: OR 1.09 [1.02–1.16])
		50.9 % Male	-Race/ethnicity	([5× week day + 2× weekend day]/7)	Attention problem: score ≥1.2 S.D.s above the mean	
		Mean age 1.8 y, S.D. 0.6 At 3 y: N = 1345 48.6% Male	-Child age -Gestational age -Family pattern and characteristics	At 1 and 3 y Completed by mother	At 7 y	
		Mean age 3.8 y, S.D. 0.6	-Cognitive stimulation, emotional support at home			
		National Longitudinal Survey of Youth (representative longitudinal data set) multicentric study of healthy children	-Urban/rural residence			
	L P	T1; N = 199	-Sex	CMU scores, based on 4 indicators:- child's average daily screen	Attentional Focusing subscale from the ECBQ-SF	CMU at T1 18 mo negatively predicted children's focused

TABLE 3 (*continued*)

Author, Year, Country	Study Design	Study Population	Covariates	Ascertainment of Exposure	Ascertainment of Outcome	Outcome(s)
				time- household background TV- maternal use of media to regulate child distress- maternal use of mobile devices while spending time with the child		attentionAt T2 22 mo: β −0.22, 95% CI, −0.35 to 0.08, *P = 0.001At T3 26 mo (through T2): β −0.12, 95% CI - 0.20 to 0.04, P = 0.003*
		60% Male	-Maternal education		Score ranging from 1 to 42	
		Mean age 17.71 mo, S.D. 0.83	-General parenting practice		At 18, 22, and 26 mo	
		T2; N = 149 Mean age = 21.11 mo, S.D. 1.04			Completed by mother	
		T3; N = 119 Mean age = 25.21 mo, S.D. 1		At 18, 22, and 26 mo Completed by mother		
		Healthy infants Study selection via Prolific, an online research platform				
Hosokawa and Katsura, 2018, Japan[47]	CS	N = 1642	Sex	Self-report questionnaire regarding children's use of mobile devices, through average use time (in minutes) on a typical day	The Japanese version of SDQ	Relative to nonregular use (<1 h/day), regular use (>1 h/d) of mobile devices was significantly linked to hyperactivity/ inattention symptoms
		51.2% Male	Family composition and social characteristics	Completed by parents	2 groups::-Abnormal group: scoring $\geq 90^{th}$ percentile-Normal group: scoring $<90^{th}$ percentile	IPTW-OR, 1.82, 95% CI, [1.15-2.87], $P < 0.01$
		Mean age: 6.88 y, S.D. 0.35	-Household income			
		Healthy children recruited from kindergartens and nursery schools in a major urban area in Japan (developmental cohort)	-Parental average spending time of talking or playing with children			
			-Children's emotional/ behavioral problems at preschool		Completed by parents	
Lin et al., 2020, Taiwan[48]	CS	N = 161	-Sex	Questionnaire on basic personal information and touch screen device usage	The CBCL 1½-5	The results of Pearson's correlation showed that usage time was related to attention problems (r 0.300, $P < 0.001$)
		53.4 % Male	-Age	Completed by parents or care giver	Completed by parents or care giver	Children who spent more time on touch screen devices were more likely to have attention problems
		Mean age 25.63 mo, S.D. 5.35	-Gestational age			
		Healthy children	-Only child,			β, 0.300; 95 % CI, 0.432-1.267; $P < 0.001$
		Monocentric study	-Primary caregiver -Parents' education			
Miller et al., 2007, Canada[46]	CS	N = 170	-Sex	Semistructured interview	Checklist with 18 ADHD behaviors listed in DSM-IV	Television viewing accounted for 2.1% (*P = 0.06) of the variance in parent ratings,* $r^{2} = 0.095$

(*continued on next page*)

TABLE 3 (*continued*)

Author, Year, Country	Study Design	Study Population	Covariates	Ascertainment of Exposure	Ascertainment of Outcome	Outcome(s)
		61.8% Male	-Age,	([5× week day + 2× weekend day]/7)	4-point scale, scores ranging from 0 to 54	
		Mean age 4.31 y, S.D. 0.51	-Socioeconomic status	Completed by parents	Completed by parents and teacher	TV viewing accounted for 5.3% ($P = 0.02$) of the variance in teacher ratings, $r^2 = 0.168$
		Healthy children recruited from community preschools as part of a longitudinal study of young children at risk for ADHD				
Ozmert et al., 2002, Turky[43]	CS	N = 689	-Sex	Auto questionnaire reported television viewing	The Turkish version of CBCL	Overall television viewing time was negatively correlated with attention problems; r, 0.2; $P < 0.01$
		49.7% Male	-Age	Confirmed with a sample who completed diaries	Raw score	Stepwise logistic regression was applied, the coefficient of regression between attention problems and watching television more than 2 h/day is OR 1.138 (CI: 1.066-1.213), $P < 0.01$
		Mean age: 7.9 y	-Social subscale	Correlation r 0.6, $P < 0.01$	Completed by parents	
		Monocentric study. Healthy children recruited from 2 primary schools		2 groups: -Watch <2 h/day -Watch >2 h/day		
Parkes et al., 2013, UK[39]	L	N = 11,014	-Family composition	Typical daily hours viewing television and playing electronic games	The SDQ	No associations were found between either type of screen time exposure >3 h/day at 5 years and change in hyperactivity/ inattention symptoms from 5 to 7 years β 0.04 (95% CI −0.10-0.18), $P = 0.581$
		48.9% Male	-Mother's characteristics	At 5 y	Change scores: age 5 years score-age 7 years scores	
		Age: 5 y, then 7 y	-Household income	Reported by mothers	Measures ranging from −10 to +10.	
		Healthy children (from the UK Millennium Cohort Study)	-Mother-child relationship at age 3	6-Point scale:-None-<1 h-1-<3 h-3-<5 h-5-<7 h-7 h or more	At 7 y	
			-Parent-child joint activities, household chaos at 5 y		Completed by mothers	
		Multicentric study	-Disability -Sleeping difficulties -Physical activity -Attitudes to school -SDQ score at 5 y			
Peralta et al., 2018, Spain[40]	L P	N = 817	-Parental education	Time spent watching TV was assessed with the question "How many hours per day does your child spend watching TV/videos?"	Conners' Parent Rating Scales	ADHD symptoms at 7 y, according to the exposures (sleep, TV, cognitively stimulating activities, and physical activity) at 4 y was not significant with an adjusted IRR 1.01 (95%

TABLE 3 (*continued*)

Author, Year, Country	Study Design	Study Population	Covariates	Ascertainment of Exposure	Ascertainment of Outcome	Outcome(s)
						CI, 0.99-1.04), $P = 0.338$
		52% Male	-Maternal smoking during pregnancy	2 groups:-≤1 h/day->1 h/day	At age 7 y	
		Age: 4 y, then 7 y From the Infancia y Medio Healthy children recruited from Ambiente birth cohort	-Duration of breastfeeding At 4 y		Completed by parents	
			-Maternal smoking habits, intelligence, mental health -Child day care/ nursery attendance before school -Child neuropsychologic conditions	At age 4 y Completed by parents		
Stevens et Muslow, 2006, US[35]	L R	2 samples of 2500 children	-Limits on watching television	-Number of hours child watched television or videos each school day and each weekend dayduring their kindergarten school year's spring	Social Rating Scale derived from the Social Skills Rating Scale	The amount of variance accounted for in symptoms of ADHD by the remaining 3 variables was 6%
		Two samples were randomly selected from nationally representative data collected from the Early Childhood Longitudinal Study Study ECLS-K	-Parent involvement -Socioeconomic status	-Type of programming -Rules for viewing experienced during the kindergarten year Completed by parents		The parameter estimate between TV exposure and symptoms of ADHD was very small and not statistically significant ($t = -1.75$)
Tamana et al., 2019, Canada[41]	L P	N = 2427 52.3 % Male Mean age: 3 y and 5 y Healthy children recruited from the CHILD birth cohort study	-Sex -Ethnicity -SES -Marital status, parenting stress, maternal depression -Attend preschool at 5 y -Sleep duration, SDB -Physical activity	Child's total screen time/day At 3 and 5 y Reported by parents 3 groups:-<30 minutes/day-30 minutes and 2 h/d->2 hours	The CBCL 1½–5 Attention problem: T score ≥65 CBCL at 5 y Completed by parents	More than 2 hours of screen time/day compared with children with less than 30 minutes of screen time/day was significantly associated with an inattention problem score OR 5.9; 95% CI, 1.6, 21.5; $P = 0.01$ More than 2 hours of screen time/day was significantly associated with an ADHD score OR 7.7; 95% CI, 1.6, 38.1, $P = 0.01$, adjusting covariates
Tomopoulos et al., 2007, US[42]	L P	N = 99	-Sociodemographic characteristics-National origin-Maternal education-Depressive symptoms-Participation in a	24-hour recall diary: media exposure at 21 and 33 mo, which included information about the duration and content of each program watched	The CBCL 1½-5	Total media exposure at 21 mo was not statistically associated with ADHD problems aOR 1.6 (CI, 1.2-2.5), $P = 0.08$

(*continued on next page*)

TABLE 3 (*continued*)

Author, Year, Country	Study Design	Study Population	Covariates	Ascertainment of Exposure	Ascertainment of Outcome	Outcome(s)
		% Male: not described	parenting program-Child temperament	At age 21 and 33 mo	Attention problem: T score ≥65	Total media exposure at 33 mo was not statistically significantly associated with ADHD problems aOR 1.6 (CI, 0.9-2.8), $P = 0.09$
		Mean age: 33.4 mo.		Completed by caregivers	At 33 mo	
		From a study of a pediatric healthy children recruited from the primary care-based parenting program called the VIP			Completed by caregiver	
Verlinden et al; , 2012, Netherland[37]	L P	N = 3913	-Sex,	Questionnaire	The Dutch version of the CBCL 1½-5	High levels of TV viewing were associated with the incidence of externalizing problems: OR, 2.00; 95% CI, 1.07-3.75; $P = 0.03$
		Children with externalizing problems	-Age	4 groups:		High levels of TV viewing were associated with persistence of preexisting externalizing problems OR, 2.59; 95% CI, 1.03-6.55; $P = 0.04$
		At 24 mo: N = 476	-National origin	-"Never or less than 0.5 hour": <0.5 h/d at 24 and 36 mo	Subscale score for externalizing problems: score ≥18	
		56.3 % Male	-Day care attendance	-"Continued low TV exposure": 0.5-1 h/d at both ages	At 18 and 36 mo	
		At 36 mo: N = 452	-Maternal and paternal age	-"Continued moderate TV exposure": 1 h/d at both ages	Completed by parents	
		Children without externalizing problems	-Educational level of the parents	-"High exposure": increase in exposure (from <0.5 h/d at 24 mo to >1 h/d at 36 mo) or continued high exposure (>0.5 h/d at 24 mo and >1 h/d at 36 mo)		
		At 24 mo n = 3437 48.4 % Male At 36 mo n = 3309	-Marital status -Monthly income -Maternal symptoms of psychiatric disorders	at 24 and 36 mo Completed by parents		
		Healthy children recruited from the generation R study	-Parenting stress -Parity			
Xie et al., 2020, China[45]	CS	N = 1897	-Socioeconomic status (family income, marital status, education of parents)	One week observation of screen time (TV/DVD watching, use tablets, computers, smartphones, and playing videos games)	The CBCL 1½-5	Preschoolers with screen time of >60 min tend to have more attention problems: t value −5.15, $P < 0.05$
		54.5 % Male	-Maternal marital status	Completed by parents	No more details found	Preschoolers with screen time of >60 min tend to have more externalizing problems: β 4.04 (CI, 1.17-6.91) $P = 0.016$

TABLE 3 (*continued*)

Author, Year, Country	Study Design	Study Population	Covariates	Ascertainment of Exposure	Ascertainment of Outcome	Outcome(s)
		Age 3-6 y Healthy children recruited in 42 kindergartens with 3842 subjects in No. 8 People's Hospital of Wuxi City (China)	-Physical activity -Sleep duration	2 groups: -< 60 min	Completed by parents	
			-Maternal depression -Primary caregivers	-> 60 min		

ADHD = Attention-deficit/hyperactivity disorder
aOR = Adjusted odds ratio
BPI = Behavioral Problems Index
CBCL = Child Behavior Checklist
CI = Confidence interval
CMU = Cumulative media use
CS = Cross-sectional
DSM-IV = Diagnostic and Statistical Manual of Mental Disorders, fourth edition
ECBQ SF = Early Childhood Behavior Questionnaire Short Form
IPTW = Inverse Probability of Treatment Weighted
L = Longitudinal
OR = Odds ratio
P = Prospective
R = Retrospective
SDB = Sleep-disordered breathing
SDQ = Strengths and Difficulties Questionnaire
SES = Socioeconomic status
VIP = Video Interaction Project

Seven of the studies included suggested that the interaction between high screen exposure and attentional difficulties was likely bidirectional.[35,38,45-48]

Discussion

This systematic review found evidence of a relationship between high exposure to screens and both immediate and long-term attentional functions in preschool children. This finding is consistent with the results of other systematic reviews that evaluated this relationship in teenagers.[50,51]

Immediate association between high screen exposure and attentional abilities at the same age

We found a positive, mild to moderate association between high levels of screen exposure and attentional difficulties. This is the first systematic review to demonstrate such a relationship in preschool children; however, these results must be interpreted with caution due to the presence of confounding factors. Although most studies took known confounding factors such as sex,[52] birth weight,[53] gestational age,[54] and socioeconomic status[55] into account, other, less-well-understood factors such as parental education, parental care, maternal primiparity, parenting stress, and child temperament were not considered.[56] Furthermore, screen exposure has also been found to be related to behavioral problems,[44-46,48] social problems,[44,45] withdrawal symptoms,[44,45] and symptoms of depression,[44] which can in turn affect attentional functions. It is therefore complex to determine the precise relationship between screen exposure and attentional functions.

The focus of the present review was on the quantity of screen exposure; however, evidence suggests that the quality of the exposure also influences attentional functions. Several studies evaluated the impact of the quality of the media exposure; for example, Chonchaiya et al. evaluated the exposure to adult TV programs and Tomopoulos et al. evaluated exposure to noneducational programs. Both studies found an association between such exposures and attentional difficulties.[41,49]

Some experimental studies provide empirical evidence that the speed of editing of a television program can affect children's attention[57] and that watching a 9-minute episode of a fast-paced television cartoon immediately impaired young children's executive functions[58] relative to watching an educational television show or drawing. The results provide evidence that screen exposure has also immediate consequences for attentional function.

Long-term impact of early screen exposure on subsequent attentional function

Six of the 10 longitudinal studies included in this review found that early screen exposure was associated with subsequent attentional difficulties. The other four studies found no relationship. The lack of unanimity in the results could be due to differences in study design and the fact that multiple contextual factors relating to media use and child development are difficult to take into account and may have confounded the results.

Screen time habits may begin as early as infancy[56]: children who watch television for more than two hours per day at age two years are more likely to watch more than two hours per day[59] and to protest more when the television is turned off at age six years[60(p6)]. Children's habits are also maintained by their immediate environment. For example, they watch TV more often if it is accessible more readily; this is particularly true for those with ADHD.[61] It would therefore seem relevant to develop programs that target the reduction of screen exposure and accessibility from infancy.

Furthermore, it has been suggested that the relationship between screen time and later behavioral outcomes might be indirect via the associated low level of active interaction between children and their caregivers.[62] Thus, the negative, long-term impact seems due to high screen exposure habits that develop in early childhood, which may be more prevalent in families with a low socioeconomic status.[63]

Reliability of measures of screen time and attentional functions

It is important to note that the evaluations used to evaluate both screen exposure and attentional function in the studies included have been found to be reliable but have some limitations. Although Ozmert et al. found a significant correlation between television viewing time reported by a questionnaire and that reported by a diary kept by parents,[45] suggesting a low level of memory bias in the ratings, and Anderson and al. found no differences between parent diaries of their child's television time and automated time-lapse video observations,[64] these measures remain subjective. Objective evaluations of screen time performed using environmental sound recording devices such as the LENA system (Language Environment Analysis System)[65] could be more accurate. Although this system is mainly used to assess the effects of television on language development,[62,66] it could be extended to studies of other aspects of screen exposure.

There is currently no single, gold-standard scale for the evaluation of attentional difficulties in preschool children. The CBCL has been shown to reliably assess attentional difficulties,[67,68] and the SDQ has also been shown to be reliable and is widely used in research,[69,70] including in studies of preschoolers.[71] The definition of ADHD, described in the fifth version of the American Psychiatric Association's Diagnostic and Statistical Manual of Mental Disorders,[72] would not be sensitive enough to detect slight and hidden attentional problems.

Direction of the relationship between media use and attentional functions in preschoolers

An important issue, which is challenging to resolve, is whether screen media use affects attentional functions or whether attentional functions affect screen media use. Several studies have addressed this question; however, the direction of the relationship has not yet been fully elucidated. In this systematic review only one study assessed the direction of the relationship between media exposure and attentional function and found a unidirectional relationship. Some studies reported that children with social-emotional difficulties are exposed to mobile technologies by their parents to calm them,[73,74] whereas others demonstrated that attention problems tend to precede pathologic gaming behaviors,[75] and that screen time predicts lower scores on developmental screening tests but not the reverse.[76] More research is therefore required to answer this important question.

What are the implications of impairing attentional functions in young children?

It is now accepted that children could learn from educational screen media; however, young children have difficulty transferring what they learn in one context to another (transfer deficit).[77] Children learn through interaction; this is especially true for language skills.[62] High levels of screen exposure could impair learning through several mechanisms. First, screens often provide attention-grabbing stimuli of low developmental value (theory of displacement), and the time spent viewing reduces the time available for developmentally appropriate learning opportunities such as playtime.[78,79] Second, screen exposure, even if it is background television, also distracts from the learning of cognitive tasks during the exposure.[80,81] Third, screen media use can alter cognitive performance through altered sleep quality.[82] A study of anatomic variations in brain structure related to level of screen exposure revealed that the integrity of brain white matter in preschoolers is altered in children whose screen use exceeds that of the AAP guidelines.[83]

The negative effects of excessive screen exposure seem to be reversible if screen time is reduced.[84] This process, called Attention Restoration Theory, was demonstrated by a systematic review in 2018.[85] However, it is our opinion that the high risks demonstrated by the results of the present study call for an avoidance of screen exposure in young children, as recommended by the AAP. Their guidelines state that technological media should be avoided before the age two years (except for video chatting), and that preschoolers should watch no more than one hour of high-quality programming per day.[86]

Limitations

This review has several limitations. Many factors may confound the relationship between screen exposure and attention difficulties, some of which are well known, whereas other, less-well-understood factors may have been underestimated in the studies included. Several of the studies included were retrospective, which may have caused bias, including memorization bias. Furthermore, social desirability bias may be present because of the general disapproval associated with excessive screen use.

Our study has several strengths. First, it is the first systematic review of the impact of screen exposure on attentional functions in young children. Second, we screened a large number of studies in several databases to ensure exhaustivity. Third, we used a rigorous methodology for the selection of studies based on the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines, and two reviewers independently selected studies. Fourth, the risk of bias was estimated using a validated scale, the NOS.

Conclusion

The results of this systematic review suggest that a high level of screen exposure is harmful for attentional functions in young children. Although conclusions regarding the relationship between screen exposure and attentional functions may be limited by many confounding factors, in view of the potential health risks of such exposure, we advocate a cautious approach to screen use. Current pediatric guidelines recommend no screen exposure before age two years and limited exposure thereafter. Considering the high proportion of children with screen exposure time that exceeds the recommended levels, further research is necessary to identify effective interventions targeting screen time reduction to create recommendations with real-life applications.

References

1. Picherot G, Cheymol J, Assathiany R, et al. Children and screens: guidelines for pediatricians and families. Arch Pediatr. 2018;25:170–174.
2. Portable devices: when you give a kid a tablet, he'll ask for more time Reportlinker. Available at: https://www.reportlinker.com/insight/give-kid-tablet-ask-time.html. Accessed September 1, 2021.
3. Definition of SCREEN TIME. Avaliable at: https://www.merriam-webster.com/dictionary/screen+time. Accessed September 1, 2021.
4. Vandewater EA, Rideout VJ, Wartella EA, Huang X, Lee JH, Shim MS. Digital childhood: electronic media and technology use among infants, toddlers, and preschoolers. Pediatrics. 2007;119:e1006–e10015.
5. Marshall SJ, Gorely T, Biddle SJH. A descriptive epidemiology of screen-based media use in youth: a review and critique. J Adolesc. 2006;29:333–349.
6. Shah RR, Fahey NM, Soni AV, Phatak AG, Nimbalkar SM. Screen time usage among preschoolers aged 2-6 in rural Western India: a cross-sectional study. J Family Med Prim Care. 2019;8:1999–2002.
7. Children and media tips from the American Academy of pediatrics. AAP.org. Avaliable at: http://www.aap.org/en-us/about-the-aap/aap-press-room/news-features-and-safety-tips/Pages/Children-and-Media-Tips.aspx. Accessed September 1, 2021.
8. Étude sur les attitudes des parents, Fondart. Avaliable at: https://www.unaf.fr/spip.php?article18412. Accessed September 1, 2021.
9. Zhu Z, Tang Y, Zhuang J, et al. Physical activity, screen viewing time, and overweight/obesity among Chinese children and adolescents: an update from

the 2017 physical activity and fitness in China-the youth study. BMC Public Health. 2019;19:197.
10. Danner FW. A national longitudinal study of the association between hours of TV viewing and the trajectory of BMI growth among US children. J Pediatr Psychol. 2008;33:1100–1107.
11. Carter B, Rees P, Hale L, Bhattacharjee D, Paradkar MS. Association between portable screen-based media device access or use and sleep outcomes: a systematic review and meta-analysis. JAMA Pediatr. 2016;170:1202–1208.
12. Hale L, Guan S. Screen time and sleep among school-aged children and adolescents: a systematic literature review. Sleep Med Rev. 2015;21:50–58.
13. Domingues-Montanari S. Clinical and psychological effects of excessive screen time on children. J Paediatr Child Health. 2017;53:333–338.
14. Gopinath B, Baur LA, Hardy LL, et al. Relationship between a range of sedentary behaviours and blood pressure during early adolescence. J Hum Hypertens. 2012;26:350–356.
15. Klesges RC, Shelton ML, Klesges LM. Effects of television on metabolic rate: potential implications for childhood obesity. Pediatrics. 1993;91:281–286.
16. Zimmerman FJ, Christakis DA. Children's television viewing and cognitive outcomes: a longitudinal analysis of national data. Arch Pediatr Adolesc Med. 2005;159:619–625.
17. Adelantado-Renau M, Moliner-Urdiales D, Cavero-Redondo I, Beltran-Valls MR, Martínez-Vizcaíno V, Álvarez-Bueno C. Association between screen media use and academic performance among children and adolescents: a systematic review and meta-analysis. JAMA Pediatr. 2019;173:1058–1067.
18. Pagani LS, Fitzpatrick C, Barnett TA, Dubow E. Prospective associations between early childhood television exposure and academic, psychosocial, and physical well-being by middle childhood. Arch Pediatr Adolesc Med. 2010;164:425–431.
19. Madigan S, McArthur BA, Anhorn C, Eirich R, Christakis DA. Associations between screen use and child language skills: a systematic review and meta-analysis. JAMA Pediatr. 2020;174:665–675.
20. Duch H, Fisher EM, Ensari I, et al. Association of screen time use and language development in hispanic toddlers: a cross-sectional and longitudinal study. Clin Pediatr (Phila). 2013;52:857–865.
21. Zimmerman FJ, Christakis DA, Meltzoff AN. Associations between media viewing and language development in children under age 2 years. J Pediatr. 2007;151:364–368.
22. Byeon H, Hong S. Relationship between television viewing and language delay in toddlers: evidence from a Korea national cross-sectional survey. PLoS One. 2015;10, e0120663.
23. Rueda MR, Posner MI, Rothbart MK. The development of executive attention: contributions to the emergence of self-regulation. Dev Neuropsychol. 2005;28:573–594.
24. Kar BR, Kar BR, eds. Cognition and brain development: Converging evidence from various methodologies. Washington, DC: American Psychological Association; 2013. xiii-328.
25. Kraybill JH, Kim-Spoon J, Bell MA. Infant attention and age 3 executive function. Yale J Biol Med. 2019;92:3–11.
26. Umiltà C, Moscovitch M, eds. Attention and performance XV. Conscious and nonconscious information processing. Cambridge, MA: The MIT Press; 1994.
27. Dehaene S. Les grands principes de l'apprentissage. Avaliable at: https://www.college-de-france.fr/site/stanislas-dehaene/symposium-2012-11-20-10h00.htm. Accessed September 1, 2021.
28. Andrade BF, Brodeur DA, Waschbusch DA, Stewart SH, McGee R. Selective and sustained attention as predictors of social problems in children with typical and disordered attention abilities. J Atten Disord. 2009;12:341–352.
29. Harpin VA. The effect of ADHD on the life of an individual, their family, and community from preschool to adult life. Arch Dis Child. 2005;90:i2–i7.
30. Daelmans B, Darmstadt GL, Lombardi J, et al. Early childhood development: the foundation of sustainable development. Lancet. 2017;389:9–11.
31. PRISMA guidelines. Avaliable at: http://www.prisma-statement.org/. Accessed September 1, 2021.
32. Media education. American Academy of pediatrics. Committee on public education. Pediatrics. 1999;104:341–343.
33. Ottawa Hospital Research Institute. Avaliable at: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp. Accessed September 1, 2021.
34. Cheng S, Maeda T, Yoichi S, Yamagata Z, Tomiwa K, Japan Children's Study Group. Early television exposure and children's behavioral and social outcomes at age 30 months. J Epidemiol. 2010;20:S482–S489.
35. Tamana SK, Ezeugwu V, Chikuma J, et al. Screen-time is associated with inattention problems in preschoolers: results from the CHILD birth cohort study. PLoS One. 2019;14, e0213995.
36. Christakis DA, Zimmerman FJ, DiGiuseppe DL, McCarty CA. Early television exposure and subsequent attentional problems in children. Pediatrics. 2004;113:708–713.
37. Gueron-Sela N, Gordon-Hacker A. Longitudinal links between media use and focused attention through toddlerhood: a cumulative risk approach. Front Psychol. 2020;11, 569222.
38. Parkes A, Sweeting H, Wight D, Henderson M. Do television and electronic games predict children's psychosocial adjustment? Longitudinal research using the UK Millennium Cohort Study. Arch Dis Child. 2013;98:341–348.
39. Peralta GP, Forns J, García de la Hera M, et al. Sleeping TV, cognitively stimulating activities, physical activity, and attention-deficit hyperactivity disorder symptom Incidence in children: a prospective study. J Dev Behav Pediatr. 2018;39:192–199.
40. Stevens T, Mulsow M. There is no meaningful relationship between television exposure and symptoms of attention-deficit/hyperactivity disorder. Pediatrics. 2006;117:665–672.
41. Tomopoulos S, Dreyer BP, Valdez P, et al. Media content and externalizing behaviors in Latino toddlers. Ambul Pediatr. 2007;7:232–238.
42. Verlinden M, Tiemeier H, Hudziak JJ, et al. Television viewing and externalizing problems in preschool children: the Generation R Study. Arch Pediatr Adolesc Med. 2012;166:919–925.
43. Modesti PA, Reboldi G, Cappuccio FP, et al. Panethnic differences in blood pressure in Europe: a systematic review and meta-analysis. PLoS One. 2016;11, e0147601.
44. Lin HP, Chen KL, Chou W, et al. Prolonged touch screen device usage is associated with emotional and behavioral problems, but not language delay, in toddlers. Infant Behav Dev. 2020;58, 101424.
45. Ozmert E, Toyran M, Yurdakök K. Behavioral correlates of television viewing in primary school children evaluated by the child behavior checklist. Arch Pediatr Adolesc Med. 2002;156:910–914.
46. Xie G, Deng Q, Cao J, Chang Q. Digital screen time and its effect on preschoolers' behavior in China: results from a cross-sectional study. Ital J Pediatr. 2020;46:9.
47. Miller CJ, Marks DJ, Miller SR, et al. Brief report: television viewing and risk for attention problems in preschool children. J Pediatr Psychol. 2007;32:448–452.
48. Hosokawa R, Katsura T. Association between mobile technology use and child adjustment in early elementary school age. PLoS One. 2018;13, e0199959.
49. Chonchaiya W, Sirachairat C, Vijakkhana N, Wilaisakditipakorn T, Pruksananonda C. Elevated background TV exposure over time increases behavioural scores of 18-month-old toddlers. Acta Paediatr. 2015;104:1039–1046.
50. Ra CK, Cho J, Stone MD, et al. Association of digital media use with subsequent symptoms of attention-deficit/hyperactivity disorder among adolescents. JAMA. 2018;320:255–263.
51. Nikkelen SWC, Valkenburg PM, Huizinga M, Bushman BJ. Media use and ADHD-related behaviors in children and adolescents: a meta-analysis. Dev Psychol. 2014;50:2228–2241.
52. Biederman J, Mick E, Faraone SV, et al. Influence of gender on attention deficit hyperactivity disorder in children referred to a psychiatric clinic. Am J Psychiatry. 2002;159:36–42.
53. Pettersson E, Sjölander A, Almqvist C, et al. Birth weight as an independent predictor of ADHD symptoms: a within-twin pair analysis. J Child Psychol Psychiatry. 2015;56:453–459.
54. Franz AP, GU Bolat, Bolat H, et al. Attention-deficit/hyperactivity disorder and very preterm/very low birth weight: a meta-analysis. Pediatrics. 2018;141, e20171645.
55. Russell AE, Ford T, Williams R, Russell G. The association between socioeconomic disadvantage and attention deficit/hyperactivity disorder (ADHD): a systematic review. Child Psychiatry Hum Dev. 2016;47:440–458.
56. Trinh MH, Sundaram R, Robinson SL, et al. Association of trajectory and covariates of children's screen media time. JAMA Pediatr. 2020;174:71–78.
57. Cooper N, Uller C, Pettifer J, Stolc F. Conditioning attentional skills: examining the effects of the pace of television editing on children's attention. Acta Paediatr. 2009;98:1651–1655.
58. Lillard A, Peterson J. The immediate impact of different types of television on young children's executive function. Pediatrics. 2011;128:644–649.
59. Certain LK, Kahn RS. Prevalence, correlates, and trajectory of television viewing among infants and toddlers. Pediatrics. 2002;109:634–642.
60. Christakis DA, Zimmerman FJ. Early television viewing is associated with protesting turning off the television at age 6. MedGenMed. 2006;8:63.
61. Lo CB, Waring ME, Pagoto SL, Lemon SC. A television in the bedroom is associated with higher weekday screen time among youth with attention deficit hyperactivity disorder (ADD/ADHD). Prev Med Rep. 2015;2:1–3.
62. Zimmerman FJ, Gilkerson J, Richards JA, et al. Teaching by listening: the importance of adult-child conversations to language development. Pediatrics. 2009;124:342–349.
63. Carson V, Spence JC, Cutumisu N, Cargill L. Association between neighborhood socioeconomic status and screen time among pre-school children: a cross-sectional study. BMC Public Health. 2010;10:367.
64. Anderson DR, Field DE, Collins PA, Lorch EP, Nathan JG. Estimates of young children's time with television: a methodological comparison of parent reports with time-lapse video home observation. Child Dev. 1985;56:1345–1357.
65. LENA. Available at: https://www.lena.org. Accessed September 1, 2021.
66. Christakis DA, Gilkerson J, Richards JA, et al. Audible television and decreased adult words, infant vocalizations, and conversational turns: a population-based study. Arch Pediatr Adolesc Med. 2009;163:554–558.
67. Oerbeck B, Overgaard KR, Pripp AH, Reichborn-Kjennerud T, Aase H, Zeiner P. Early predictors of ADHD: evidence from a prospective birth cohort. J Atten Disord. 2020;24:1685–1692.
68. Aebi M, Winkler Metzke C, Steinhausen HC. Accuracy of the DSM-oriented attention problem scale of the child behavior checklist in diagnosing attention-deficit hyperactivity disorder. J Atten Disord. 2010;13:454–463.
69. Hall C, Guo B, Valentine A, et al. The validity of the Strengths and Difficulties Questionnaire (SDQ) for children with ADHD symptoms. PLoS One. 2019;14, e0218518.
70. Croft S, Stride C, Maughan B, Rowe R. Validity of the strengths and difficulties questionnaire in preschool-aged children. Pediatrics. 2015;135:e1210–e1219.
71. Rimvall MK, Elberling H, Rask CU, Helenius D, Skovgaard AM, Jeppesen P. Predicting ADHD in school age when using the Strengths and

Difficulties Questionnaire in preschool age: a longitudinal general population study, CCC2000. Eur Child Adolesc Psychiatry. 2014;23:1051–1060.
72. Moran M. APA advocacy Wins Coverage of *DSM* Codes in 12 states, D.C. Psychiatr News. 2016;51:1.
73. Radesky JS, Peacock-Chambers E, Zuckerman B, Silverstein M. Use of mobile technology to calm upset children: associations with social-emotional development. JAMA Pediatr. 2016;170:397–399.
74. Kostyrka-Allchorne K, Cooper NR, Simpson A. The relationship between television exposure and children's cognition and behaviour: a systematic review. Dev Rev. 2017;44:19–58.
75. Ferguson CJ, Ceranoglu TA. Attention problems and pathological gaming: resolving the "chicken and egg" in a prospective analysis. Psychiatr Q. 2014;85:103–110.
76. Madigan S, Browne D, Racine N, Mori C, Tough S. Association between screen time and children's performance on a developmental screening test. JAMA Pediatr. 2019;173:244–250.
77. Choi K, Kirkorian HL, Pempek TA. Understanding the transfer deficit: contextual mismatch, proactive interference, and working memory affect toddlers' video-based transfer. Child Dev. 2018;89:1378–1393.
78. Kopp CB. Regulation of distress and negative emotions: a developmental view. Dev Psychol. 1989;25:343–354.
79. Neuman SB. The displacement effect: assessing the relation between television viewing and reading performance. Read Res Q. 1988;23:414–440.
80. Wyss NM, Kannass KN, Haden CA. The effects of distraction on cognitive task performance during toddlerhood. Infancy. 2013;18:604–628.
81. Erickson LC, Newman RS. Influences of background noise on infants and children. Curr Dir Psychol Sci. 2017;26:451–457.
82. Adelantado-Renau M, Diez-Fernandez A, Beltran-Valls MR, Soriano-Maldonado A, Moliner-Urdiales D. The effect of sleep quality on academic performance is mediated by internet use time: DADOS study. J Pediatr. 2019;95:410–418.
83. Hutton JS, Dudley J, Horowitz-Kraus T, DeWitt T, Holland SK. Associations between screen-based media use and brain white matter integrity in preschool-aged children. JAMA Pediatr. 2019;4, e193869.
84. Mistry KB, Minkovitz CS, Strobino DM, Borzekowski DLG. Children's television exposure and behavioral and social outcomes at 5.5 years: does timing of exposure matter? Pediatrics. 2007;120:762–769.
85. Ohly H, White MP, Wheeler BW, et al. Attention restoration theory: a systematic review of the attention restoration potential of exposure to natural environments. J Toxicol Environ Health B Crit Rev. 2016;19:305–343.
86. Media use in childhood: evidence-based recommendations for caregivers. Available at: https://www.apa.org/pi/families/resources/newsletter/2019/05/media-use-childhood. Accessed September 1, 2021.